

Chapter 8 – Package Management & System Updates

Learning Objectives

After completing this chapter, you will be able to:

- Understand what a package manager is.
 - Learn how the APT package manager works.
 - Install, update, upgrade, and remove software.
 - Search for packages.
 - View installed packages.
 - Clean unnecessary files.
 - Troubleshoot common APT errors.
 - Keep Kali Linux secure by applying updates regularly.
-

8.1 What is a Package Manager?

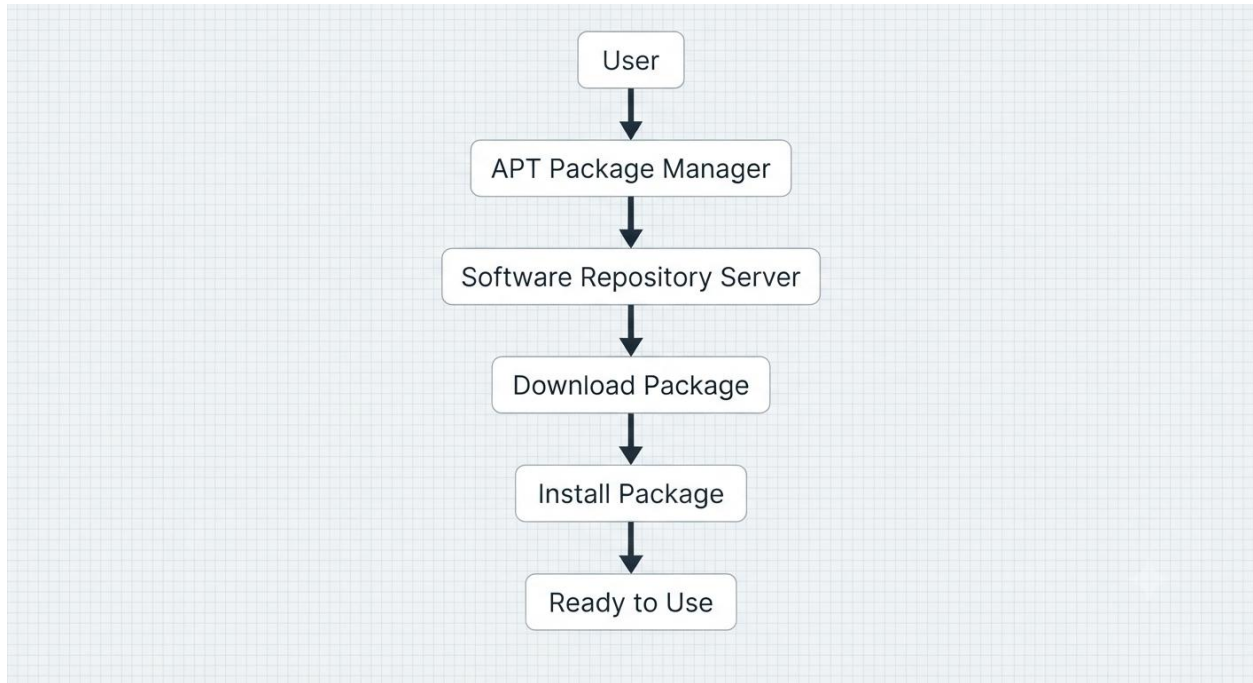
A **Package Manager** is a tool that installs, updates, removes, and manages software on Linux.

Instead of downloading applications from random websites, Linux uses trusted software repositories.

This makes software installation:

- Easier
 - Faster
 - More secure
 - Easier to maintain
-

Package Management Workflow



8.2 What is APT?

APT stands for **Advanced Package Tool**.

APT is the default package manager for Debian-based operating systems such as:

- Debian
- Ubuntu
- Kali Linux

APT automatically:

- Downloads packages
 - Resolves dependencies
 - Installs software
 - Removes software
 - Updates software
-

8.3 Updating the Package List

Before installing software, refresh the package database.

Run:

```
sudo apt update
```

Command Breakdown

Command	Purpose
sudo	Administrative privileges
apt	Package Manager
update	Refresh package list

8.4 Upgrading Installed Packages

Install available updates.

```
sudo apt full-upgrade -y
```

Why Use full-upgrade?

Kali Linux is a rolling release distribution.

Unlike Ubuntu LTS, Kali frequently changes packages and dependencies.

full-upgrade safely handles:

- New packages
 - Dependency changes
 - Obsolete packages
-

8.5 Installing Software

Install new software.

Example:

```
sudo apt install git
```

Install multiple packages:

```
sudo apt install git curl wget nano vim -y
```

Command Breakdown

sudo apt install git

Part	Description
sudo	Run as Administrator
apt	Package Manager
install	Install Software
git	Package Name

8.6 Searching for Packages

Not sure if a package exists?

Search:

```
apt search wireshark
```

Example Output

wireshark

Network protocol analyzer

8.7 Viewing Installed Packages

Display installed packages.

```
apt list --installed
```

You can also search:

```
apt list --installed | grep git
```

8.8 Removing Software

Remove software.

```
sudo apt remove git
```

This removes the application but keeps configuration files.

8.9 Purging Software

To completely remove a package including configuration files:

```
sudo apt purge git
```

Difference:

remove	purge
Removes Program	Removes Program
Keeps Config Files	Removes Config Files

8.10 Removing Unused Packages

Over time, unused dependencies accumulate.

Remove them using:

```
sudo apt autoremove
```

This frees storage space.

8.11 Cleaning the Package Cache

Downloaded packages are stored locally.

Clean them:

```
sudo apt clean
```

Remove only outdated packages:

```
sudo apt autoclean
```

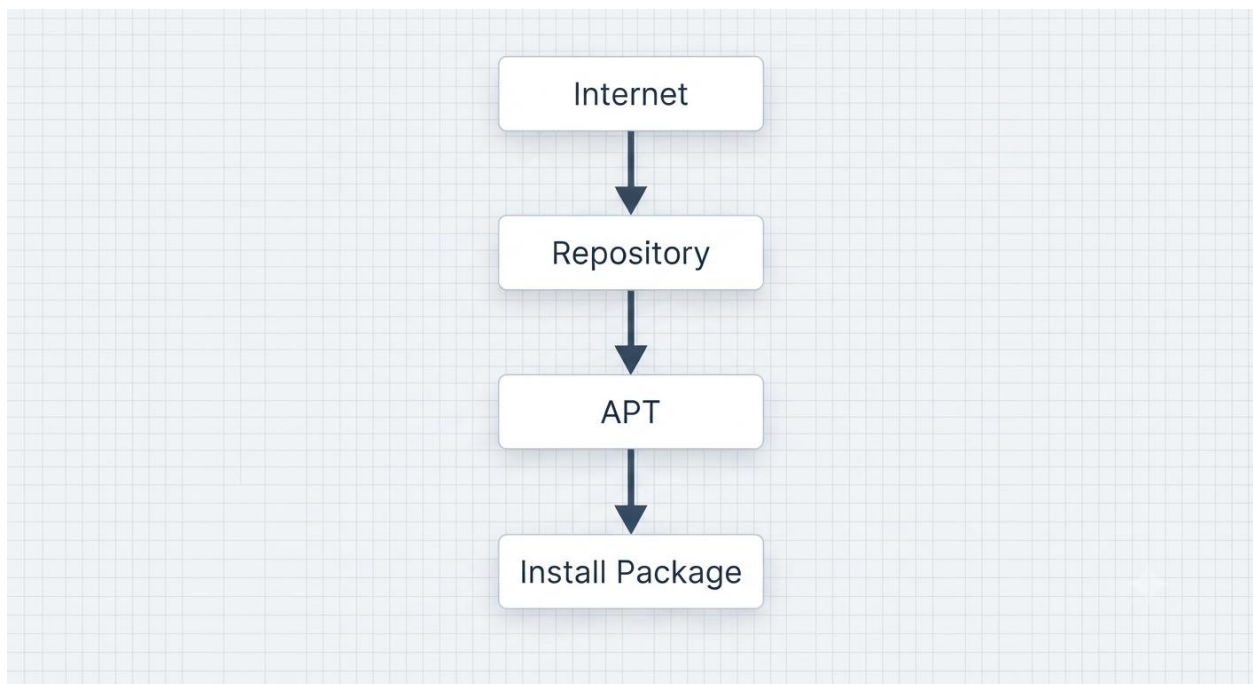
Difference

Command	Purpose
clean	Remove all cached packages
autoclean	Remove only outdated packages

8.12 Understanding Repositories

Repositories are trusted servers that store software packages.

Instead of downloading software manually:



Kali Repository

Repository configuration file:

/etc/apt/sources.list

View it:

```
cat /etc/apt/sources.list
```

8.13 Checking Package Information

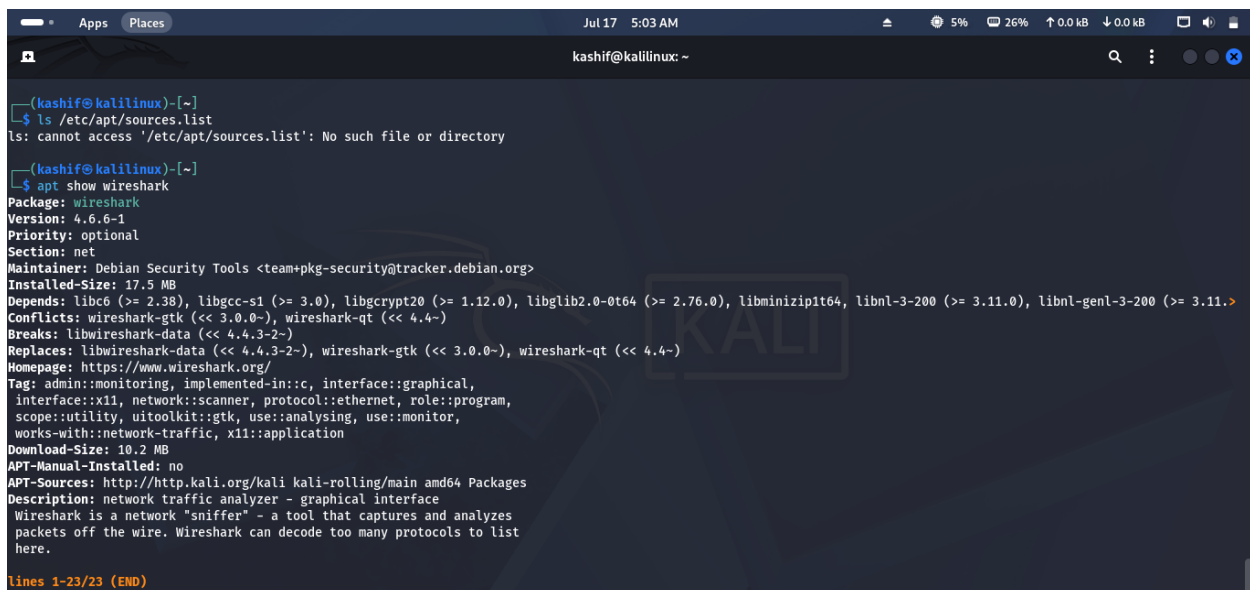
View package details.

Example:

```
apt show wireshark
```

Displays:

- Version
- Maintainer
- Description
- Dependencies
- Download Size



```
(kashif@kalilinux)-[~]
$ ls /etc/apt/sources.list
ls: cannot access '/etc/apt/sources.list': No such file or directory

(kashif@kalilinux)-[~]
$ apt show wireshark
Package: wireshark
Version: 4.6.6-1
Priority: optional
Section: net
Maintainer: Debian Security Tools <team+pkg-security@tracker.debian.org>
Installed-Size: 17.5 MB
Depends: libc6 (>= 2.38), libgcc-s1 (>= 3.0), libgcrypt20 (>= 1.12.0), libglib2.0-0t64 (>= 2.76.0), libminizip1t64, libnl-3-200 (>= 3.11.0), libnl-genl-3-200 (>= 3.11.0)
Conflicts: wireshark-gtk (< 3.0.0-), wireshark-qt (< 4.4~)
Breaks: libwireshark-data (< 4.4.3-2~)
Replaces: libwireshark-data (< 4.4.3-2~), wireshark-gtk (< 3.0.0-), wireshark-qt (< 4.4~)
Homepage: https://www.wireshark.org/
Tag: admin::monitoring, implemented-in::c, interface::graphical,
interface::x11, network::scanner, protocol::ethernet, role::program,
scope::utility, uitoolkit::gtk, use::analysing, use::monitor,
works-with::network-traffic, x11::application
Download-Size: 10.2 MB
APT-Manual-Installed: no
APT-Sources: http://http.kali.org/kali kali-rolling/main amd64 Packages
Description: network traffic analyzer - graphical interface
Wireshark is a network "sniffer" - a tool that captures and analyzes
packets off the wire. Wireshark can decode too many protocols to list
here.

lines 1-23/23 (END)
```

8.14 Downloading Without Installing

Sometimes you only need the package file.

`apt download nmap`

Useful for:

- Offline installation
- Package analysis

8.15 Keeping Kali Updated

Security tools change frequently.

Recommended schedule:

Task	Frequency
apt update	Daily
full-upgrade	Weekly
autoremove	Monthly
clean	Monthly

8.16 Common APT Errors

Error

Unable to locate package

Solution:

`sudo apt update`

Error

Failed to fetch

Possible Causes

- No Internet
 - Repository unavailable
-

Error

Could not get lock

Another APT process is running.

Wait until it finishes.

Error

Broken packages

Repair:

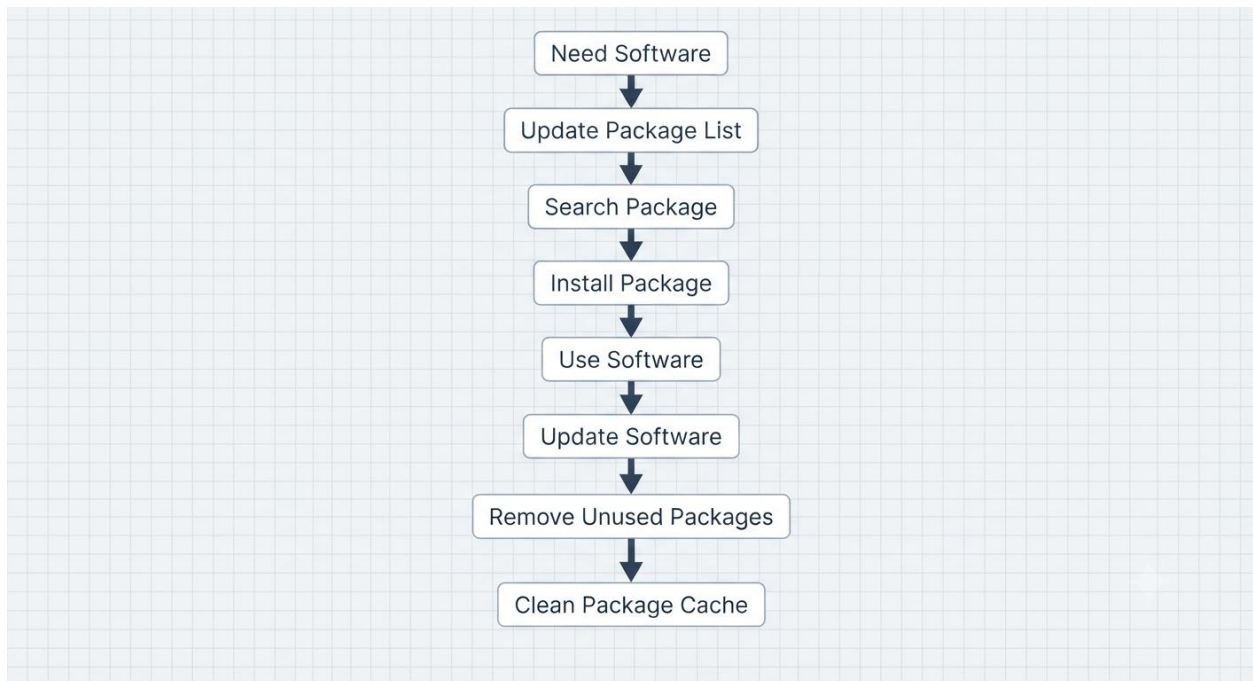
```
sudo apt --fix-broken install
```

8.17 Useful APT Commands

Command	Purpose
apt update	Refresh package list
apt full-upgrade	Upgrade system
apt install	Install package
apt remove	Remove package
apt purge	Remove package + configuration
apt search	Search packages
apt show	Show package information

apt autoremove	Remove unused packages
apt clean	Clear cache
apt autoclean	Remove old cache
apt download	Download package

8.18 Complete Package Management Workflow



8.19 Best Practices

- Always run apt update before installing software.
- Regularly apply security updates.
- Install software only from trusted repositories.
- Remove unused packages periodically.
- Avoid downloading software from unknown websites.
- Read package descriptions before installation.
- Keep enough free disk space for updates.

Chapter Summary

In this chapter, you learned:

- What a package manager is.
- How APT works.
- How to install, update, upgrade, and remove software.
- How to search for packages.
- How repositories function.
- How to troubleshoot common package management errors.
- How to maintain Kali Linux using best practices.

One-Line Summary

APT simplifies software management in Kali Linux by securely installing, updating, and maintaining packages through trusted repositories, ensuring the operating system remains current and reliable.
